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VECTOR DOT PRODUCT AND DIRECTION COSINES

DOT PRODUCT PROPERTIES

- The dot product of two vectors u and v is defined by the formula: $u \cdot v = |u| |v| \cos \theta$.
- If the angle θ between the vectors is 90 degrees, then the dot product $u \cdot v = 0$.
- If the angle θ between the vectors is 0 degrees, then the dot product $u \cdot v = |u| |v|$.

DIRECTION COSINES AND DIRECTION ANGLES

Consider a vector V represented by the components (v_1, v_2, v_3) in a three-dimensional coordinate system. The direction angles (α , β , and γ) are the angles that the vector V makes with the positive x , y , and z axes respectively.

The direction cosines for these angles are defined as:

- $\cos \alpha = v_1 / |V|$
- $\cos \beta = v_2 / |V|$
- $\cos \gamma = v_3 / |V|$

In these equations, $|V|$ represents the magnitude or length of the vector V . These values describe the orientation of the vector in space relative to the coordinate axes.